

CHRONIC HEPATITIS C

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INTRODUCTION

Hepatitis C is an inflammation of the liver caused by the hepatitis C virus (HCV).

Hepatitis C virus is primarily a blood-borne virus and the most common modes of infection are through;

1. Injecting drug through sharing of injection equipment.
2. Inadequate sterilization of medical equipment.
3. Transfusion of unscreened blood and blood products.
4. Unsafe sex practices that lead to exposure to blood, people with multiple sexual partners and among men who have sex with men. However, these modes of transmission are less common.

Acute Hepatitis C infections are usually asymptomatic. This results in significant challenges for diagnosis and surveillance. Around 30% of infected person clear the virus spontaneously within 6 months of infection without treatment. The remaining 70 % will develop chronic HCV infection. Amongst these group, the risk of cirrhosis ranges from 15-30 % within 20 years.¹

Chronic infection with Hepatitis C virus (HCV) often lead to complications such as liver cirrhosis which can lead to liver failure, hepatocellular carcinoma and early death.

Unlike Hepatitis B, there is currently no effective vaccine against hepatitis C.

However, Direct-acting antiviral agents (DAAs) can cure more than 95% of persons affected by HCV.

In Singapore, **acute hepatitis C infection** is a notifiable disease under section 6 of the infectious disease act within 72 hours from time of diagnosis.² **Chronic hepatitis C need not be reported to MOH (Refer to [MOH circular 2019](#) in the intranet for notification criteria).**

Epidemiology

Globally, an estimated 50 million people have chronic Hepatitis C virus infection. There are approximately 1 million new infections occurring per year. It is a leading cause of liver disease and death.³

In Singapore, the seroprevalence is low (0.37 – 0.54 %) based on blood donor studies. Majority are among those with history of injecting drug use (IDU).^{4,5}

Based on local study, approximately 45% of people who inject drugs show evidence of current or past HCV infection.⁶

SCREENING AND DIAGNOSIS

Screening

Referring to Report of the Screening Test Review Committee released in March 2019, screening is recommended for:

1) At Risk Groups:

- a) Children born to Hepatitis C Virus (HCV) positive mothers
- b) Chronic haemodialysis patients
- c) Past or present intravenous drug abusers
- d) Individuals who underwent invasive procedures in health-care facilities with inadequate infection control practices
- e) Individuals with known exposures to HCV, e.g. healthcare workers following needle stick injury involving HCV-positive blood, or recipients of blood or organs from a donor who tested HCV-positive
- f) Individuals whose past or present sex partners were/are HCV infected or intravenous drug abusers
- g) HIV patients

2) Healthcare Workers:

- a) All healthcare workers (HCW) with direct patient contact are encouraged to have their status checked.
- b) HCW who are practising in specialties or areas involving exposure-prone procedures[†].

Diagnosis

Diagnosis depends on the detection of antibodies to recombinant antigens (Anti-HCV antibody) and detection of viral RNA (e.g., by PCR techniques).

[†] Exposure-prone procedures are invasive procedures where injury may result from exposure of the patient's open tissues to the blood of the HCW.

Initial Assessment

Clinical Presentation

Most patients do not have symptoms in the first week of infection. Symptoms may develop anywhere between 2 weeks to 6 months.

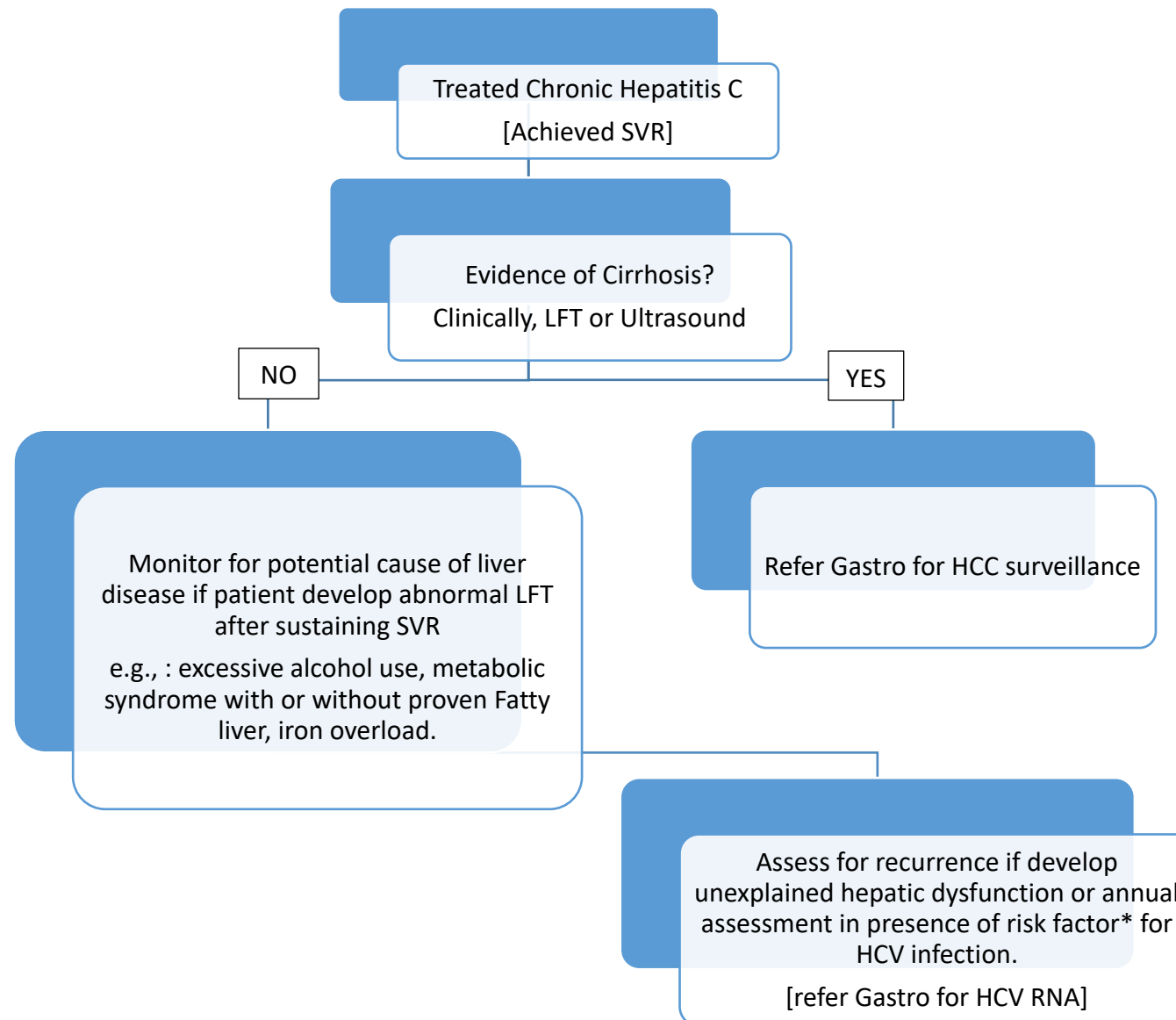
In the local context, most patients in the primary care setting were either discharged from specialist or defaulted follow up.

Patients who presented with typical symptoms of acute viral hepatitis such as fever, lethargy, anorexia, abdominal pain, jaundice, pale stools, or tea coloured urine should be offered testing. Decision to screen for Hepatitis C should be contextualized in the clinical context in correlation with management and disposition plans.

Patients who are treated and discharged from hospital would have achieved sustained virological response (SVR) defined by undetectable HCV RNA $\geq$ 12 weeks after treatment completion with direct-acting antiviral (DAA) and transaminase normalization.

Likelihood of achieving SVR with DAA generally exceeds 95%.

MANAGEMENT AND FOLLOW-UP



** People who achieve SVR can have HCV recurrence due to reinfection or late relapse. Annual testing for HCV reinfection for patients with ongoing risk for HCV infection such as injection drug use or high-risk sexual exposure.*

Patients who are discharged generally does not require follow up if there are no evidence of cirrhosis.

Assessment for other causes of liver disease is recommended for patients with persistently elevated transaminase levels after SVR.

Patients who have risk of recurrence [habit of adding tattoo, IVDU, multiple sexual partners] should be counselled.

Cirrhotic patients are at risk for Hepatocellular Carcinoma and should undergo surveillance every 6 months with ultrasound [with or without AFP testing] with Gastro SOC.

If relapse is suspected or cannot be ruled out, refer to Gastro for HCV-RNA testing.

HCV antibody remains positive in most patients after achieving SVR, testing for recurrence via HCV RNA is recommended.

Non-Pharmacological / Lifestyle

Patient should be educated on modes of HCV transmission and means to prevent it.

HCV-infected persons should be counselled on:

1. Avoid sharing toothbrushes, and dental or shaving equipment. Cover any bleeding wound to prevent others coming into contact with their blood.
2. Person who inject drugs are at heightened risk of reinfection; counsel to avoid reusing or sharing syringes, needles, water, cotton and other drug preparation equipment.
3. Advised not to donate blood and to discuss HCV serostatus prior to donation of body organs, other tissue or semen.
4. Persons with HIV & those with multiple sexual partners should be reinforced to use barrier precautions.
5. Household surfaces contaminated with visible blood from HCV- infected person should be cleaned using dilution of 1 part household bleach to 9 parts water. Gloves should be worn when cleaning up blood spills.

Screening for at risk family members such as children of person with HCV infection and sexual contacts is recommended.

HCV is not spread by sneezing, hugging, holding hands, coughing, sharing eating utensils or drinking glasses nor through food or water.

Pharmacological / Medications

Direct-acting antiviral agents prescribed by a specialist physician (gastroenterologist, hepatologist, of infectious disease specialist).

Self-Monitoring

Patients who have achieved SVR can develop recurrence from reinfection or relapse. They should be advised to monitor for:

1. Jaundice.
2. Abdominal Pain
3. Pale stools or tea coloured urine
4. Loss of weight and/or loss of appetite.
5. Vomiting of blood
6. Per rectal bleeding.
7. Abdominal swelling

Detection & Treatment of Complications

Signs or symptoms suggesting decompensated liver disease, cirrhosis or hepatocellular carcinoma should be referred to Gastroenterologist for further management.

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